Transverse phase of the complete dS_3 crossing kernel: a punctured-window expansion and logarithmic scaling

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Working research note — 31 October 2026

Abstract

The physical normalization phase changes the logarithmic transverse scaling inferred from one Omega factor alone. Using the established Wilson reflection and the two-series LPRS formula, we obtain a phase-shifted sine kernel away from the collision surface and a pointwise sinc limit for the complete scalar matrix element. An explicit finite-mass bound retains the whole companion series. Neither result establishes a distributional limit or the heavy-mass limit of a physical wave packet.

1 Model, coordinates and claims

The radius is one and $G_N = 0$. We keep proper $SO^+(3,1)$, BD scalar observers, KG matter, a fixed positive observer-label lower bound, and half-period regulated creation blocks. U_{00} is the scalar matrix element in the physical CG-period convention, not a separate unitary scalar transform. The principal-series label M corresponds to static energy $\sqrt{1+M^2}$.

Fix real u, v, positive p, q, and put

$$c = M, \quad b = M + u, \quad a = M + u + v, \quad d = M + v + r, \qquad z = -ir/2,$$

$$A_j = \tfrac{1}{2} + i\alpha_j, \qquad (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \tfrac{1}{2}(v + q, p - u, -u - p, v - q).$$

Our real transfer window requires $|\alpha_j| \le 1/2$ and $|\alpha_j + r/2| \le 1/2$. Define a continuous phase, with $\delta(0) = 0$, by

$$\delta(r) = \Im\left[2\log\Gamma(1 - ir/2) + \sum_{j=1}^{4}\{\log\Gamma(A_j + ir/2) - \log\Gamma(A_j)\}\right]. \tag{1}$$

For fixed $r \neq 0$, uniformly on compact subsets of this punctured window,

$$U_{00} = \frac{\sin(r\log M - \delta(r))}{\pi r} + O(M^{-1}). \tag{2}$$

For transfers strictly inside the $r = 0$ window, each fixed real λ instead gives

$$\lim_{M\to\infty} \frac{\pi U_{00}(r = \lambda/\log M)}{\log M} = \frac{\sin\lambda}{\lambda}, \qquad \left.\frac{\sin\lambda}{\lambda}\right|_{\lambda=0} = 1. \tag{3}$$

This is a pointwise scaling statement. The normalization phase must be included: the single collision factor had the different profile $\sin(\lambda/2)/(\lambda/2)$.

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2 The two oscillatory branches and the physical phase

The established reflection identity yields $U_{00} = 2\Re(C_+XY)$, where $X = \Omega(h, h')$ and $Y = \Omega(1-h, h')$, $h = (1+ib)/2$, $h' = (1+id)/2$. The two-series formula [1], with the Wilson reflection [2], has the useful exact structure

$$X = c_A F + c_B G, \qquad Y = y_A F + y_B H.$$

Here F is the same series in both expressions:

$$F = {}_4F_3(A - z;\, 1 + id,\, 1 - ib - z,\, 1 - z;\, 1),$$

$$H = {}_4F_3(A;\, 1 + i(M + v) - z,\, 1 - ib,\, 1 + z;\, 1),$$

where $A - z$ denotes all four numerator parameters. The large-parameter series is

$$G = {}_4F_3(A + ib;\, 1 + i(b + M + v + r/2),\, 1 + ib,\, 1 + i(b - r/2);\, 1).$$

The coefficients of Y are

$$y_A = \Gamma(z)\frac{\Gamma(1 - ib)}{\Gamma(1 - ib - z)}\frac{\Gamma(A_3 - z)\Gamma(A_4 - z)}{\Gamma(A_3)\Gamma(A_4)},$$
$$y_B = \Gamma(-z)\frac{\Gamma(1 + id)}{\Gamma(1 + i(M + v) - z)}\frac{\Gamma(A_1)\Gamma(A_2)}{\Gamma(A_1 - z)\Gamma(A_2 - z)}.$$

Let $D_0 = C_+c_A$, $E_0 = C_+c_B$ and define

$$B_s = \frac{\Gamma(A_1)\Gamma(A_2)}{\Gamma(A_1 - z)\Gamma(A_2 - z)}, \quad A_s = \frac{\Gamma(A_3 - z)\Gamma(A_4 - z)}{\Gamma(A_3)\Gamma(A_4)}.$$

Substitution of the physical three-sphere periods gives

$$D_0 = D_\infty M^z[1 + O(M^{-1})],$$
$$|D_\infty| = \frac{e^{\pi r/4}}{4\pi|\Gamma(1 - z)|}\sqrt{\prod_{j=3,4}\frac{\cosh\pi(\alpha_j + r/2)}{\cosh\pi\alpha_j}},$$
$$\arg D_\infty = -\arg B_s + \arg\Gamma(1 + z). \tag{4}$$

The phases here are continuous, not principal-value fits. The Gamma argument/weight certificate is saved in the audit. Shifted Stirling expansion [3] gives the large-argument phase explicitly as $-r\log M/2 - r(3r + 4v)/(8M) + O(M^{-2})$; the remaining terms give the phase in (4). The modulus follows from $|\Gamma(1/2 + ix)|^2 = \pi/\cosh\pi x$ and $|\Gamma(1 + ix)|^2 = \pi x/\sinh\pi x$.

Gamma ratios imply $y_A \sim \Gamma(z)A_s(-iM)^z$ and $y_B \sim \Gamma(-z)B_s(iM)^{-z}$. Thus

$$D_0(y_A + y_B) = T(r)e^{-ir\log M} + J(r) + O(M^{-1}),$$

$$T = D_\infty\Gamma(z)e^{-i\pi z/2}A_s = \frac{i\,e^{i\delta(r)}}{2\pi r}, \qquad J = D_\infty\Gamma(-z)e^{-i\pi z/2}B_s \in i\mathbb{R}.$$

The nonoscillating branch has zero real part at leading order. Its cancellation is algebraic: $\Gamma(-z)\Gamma(1 + z) = z|\Gamma(z)|^2$ is imaginary. The cosh factors cancel in $|T|$, leaving $1/(2\pi|r|)$. Taking twice the real part proves the leading expression in (2), once the series errors are controlled.

3 A finite-mass bound and the limits it supports

Put $S = y_A + y_B$, $m_A = \min(d, b - r/2) > 0$, $m_B = \min(M + v + r/2, b) > 0$. The small-numerator coefficient comparison used in the collision analysis gives

$$|F - 1| \le R_F = \frac{\zeta(2) - 1}{1 + m_A^2}, \qquad |Y - S| \le R_Y = (\zeta(2) - 1)\left(\frac{|y_A|}{1 + m_A^2} + \frac{|y_B|}{1 + m_B^2}\right).$$

These are exact-arithmetic analytic inequalities; the finite numerical evaluation of their right-hand sides is not interval certified.

For completeness the whole G series, not merely its first term, is bounded as follows. Let

$$B = b - |r|/2 > 0, \qquad A_* = \max_j |\alpha_j| + |r|/2, \qquad b + M + v + r/2 \ge B.$$

For $k = n + 1$ its term ratio obeys

$$|g_{n+1}/g_n| \le \sqrt{1 + B^2/k^2}\left(1 + \frac{2BA_* + A_*^2}{k^2 + B^2}\right)^2.$$

The sinh product [4] bounds every term by

$$B_b = \sqrt{\frac{\sinh \pi B}{\pi B}}\, e^{2\pi A_* + \pi A_*^2/B}.$$

The exact squared-ratio comparison gives a monic polynomial $P(x) = x^{10} + \sum_{j<10} p_j x^j$. Set $N_* = 2 + \lceil \max_{j<10} |p_j| \rceil$. All translated coefficients at N_* are positive by the monic polynomial root bound, also applied to its normalized derivatives. Consequently the term ratio is at most $((n+1)/(n+2))^{3/2}$ for $n \ge N_*$, and

$$|G| \le B_G = (3N_* + 3)B_b.$$

Here $N_* = O(M^8)$ is an analytic threshold, not a numerical loop length. The Gamma prefactors give $|E_0| = O(M^{-1/2} e^{-\pi M})$ on compact transfers, so $|E_0| B_G = O(M^8 e^{-\pi M/2})$ suffices.

Combining these estimates yields the explicit bound

$$|U_{00} - 2\Re(D_0 S)| \le 2|D_0|\{R_Y + R_F(|S| + R_Y)\} + 2|E_0| B_G(|S| + R_Y). \tag{5}$$

At fixed nonzero r, this error is $O(M^{-2})$ plus an exponentially small term. The Gamma coefficient expansion then proves (2).

For fixed $\lambda \ne 0$, $r = \lambda / \log M$ gives $|y_A| + |y_B| = O(\log M)$, while the coefficient-ratio expansions remain uniform for the small shifts. Their errors are $O(\log M / M)$, and (5) contributes $O(\log M / M^2)$ plus an exponential term. Also $\delta(r) = O(r)$; division by $\log M$ proves (3). The $\lambda = 0$ case uses the preceding peak theorem separately.

In fact

$$\delta'(0) = -\tfrac{1}{2}\Re\{2\psi(1) - \sum_j \psi(A_j)\}.$$

The removable $r = 0$ value of the leading sine expression therefore agrees with the logarithm and constant of the peak theorem. This matching is not a proof of uniform error across $r = 0$: the separate-branch bound contains $1/|r|$.

4 Numerical checks, preserved failures and next task

We use $(u, v, p, q) = (0.1, -0.1, 0.4, 0.4)$, $M = 16, 64, 256, 1024$, and both signs of either $r = 0.08$ or $r = 1/\log M$. Sixteen complete reflected-kernel evaluations use 90 decimal digits and at least 128 terms per series. All obey (5), allowing separately for the saved truncation tail. For positive $r = 0.08$:

M	error vs. $2\Re(D_0 S)$	bound (5)	error vs. sine form
16	$8.41\,10^{-4}$	$9.22\,10^{-1}$	$5.58\,10^{-3}$
64	$6.74\,10^{-5}$	$1.59\,10^{-3}$	$1.47\,10^{-3}$
256	$5.06\,10^{-6}$	$1.03\,10^{-4}$	$4.09\,10^{-4}$
1024	$3.64\,10^{-7}$	$6.69\,10^{-6}$	$1.14\,10^{-4}$

The explicit finite-mass bound controls a Gamma-exact approximation, not the final sine expression. These errors must not be conflated.

The sinc convergence is slow and not monotone in this range. For $\lambda = 1$, $\pi U_{00}/\log M$ is approximately $0.9225, 0.9975, 0.9977, 0.9853$; the limiting value is $\sin 1 \simeq 0.84147$. The phase correction explains why peak-shape plots at moderate mass would be misleading.

An independent Decimal/Fraction replay and symbolic Gamma-phase check passes 203 checks. Four selected points were recomputed at 120 digits and doubled cutoff. At $M = 16$ the relative changes are about $1.63\,10^{-23}$ and $9.29\,10^{-24}$; at the two $M = 1024$ scaled points they are about $1.52\,10^{-86}$ and $1.42\,10^{-86}$. These checks do not establish interval rounding bounds or accuracy at unsampled parameters.

The earlier collision parameter-derivative job has now exited normally. Its twelve 100-digit comparisons and subsequent saved-arithmetic audit pass. The earlier prototype coefficient bug, failed large-mass acceleration, shadow amplification, and insufficient-pi audit are all retained. Their old PDFs describe their historical checkpoints; new completion records give the current verification status.

The next analytic task is a bound uniform through the transverse collision, followed by oscillatory-tail control against the actual creation amplitudes and spectral measures. Neither (2) nor (3) alone permits a delta replacement in a Bose four-point function. Full Fock domains, finite-G_N dynamics and a microscopic de Sitter holographic dictionary remain open. The results here use known special-function formulas in a specified model; no priority claim is made.

References

[1] J. Liu et al., *d-dimensional SYK, AdS Loops, and 6j Symbols*, arXiv:1808.00612v3, Eqs. (3.35)–(3.38).

[2] W. Groenevelt, *The Wilson function transform*, arXiv:math/0306424v1, Proposition 4.4 and Remark 4.5. No Wilson-measure unitarity is assumed.

[3] NIST DLMF, Section 5.11, shifted Stirling and Gamma-ratio asymptotics with sector restrictions.

[4] NIST DLMF, Section 4.36.1, sinh infinite product. Sources checked 31 October 2026.